

Multiple Choice (10 marks)

1. A digital signature needs a
 - (a) Private-key system
 - (b) Shared-key system
 - (c) Public-key system
 - (d) All of them
2. Public key encryption is advantageous over Symmetric key Cryptography because of
 - (a) Speed
 - (b) Space
 - (c) Key exchange
 - (d) Key length
3. _____ is a popular tool used for network analysis in multiprotocol diverse network.
 - (a) Snort
 - (b) SuperScan
 - (c) Burp Suit
 - (d) EtterPeak
4. What are the port states determined by Nmap?
 - (a) Active, inactive, standby
 - (b) Open, half-open, closed
 - (c) Open, filtered, unfiltered
 - (d) Active, closed, unused
5. A proxy rewall lters at the
 - (a) Physical layer
 - (b) Application layer
 - (c) Data link layer
 - (d) Network layer

True / False (10 marks)

Answer True or False

6. True or False: Nessus is primarily designed for network intrusion detection and real-time traffic analysis.
7. True or False: Escaping queries is an effective method to prevent SQL injection attacks by neutralizing malicious input.
8. True or False: The sub key length at each round of DES is 56 bits, derived from the original 64-bit key after discarding parity bits.
9. True or False: The Heartbleed bug allows an attacker to read data from memory outside the bounds of a buffer.
10. True or False: A Base Signal Station is primarily used for data transmission in mobile networks, similar to a Base Transceiver Station.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the primary objectives of authentication in cybersecurity, focusing on its role in user identity verification and the implications for system security.
12. Discuss the implications of buffer-overflow vulnerabilities in application development, including their potential exploitation by attackers and strategies for mitigation.

End of Paper